

Decoding IPOs: Analysing the Indian Primary Market as an Emerging Asset Class

Project overview

Build a production-ready web platform that treats Indian IPOs as an independent asset class. The site will host an interactive IPO analytics dashboard, research summaries (whitepaper), predictive/scoring tools, and downloadable datasets. The work is split into discrete parts so teams can work in parallel: Web development, Data collection & engineering, Data analysis & modelling with Visualisation.

Workstreams

1) Web Development — (Website + Dashboard UI/UX)

Goal: Build a responsive website that hosts the analytics dashboard, report pages, model tools, and dataset downloads.

Key pages/modules

- Landing / Overview (project summary + key metrics snapshot)
- Interactive Dashboard (embedded visualisations + filters)
- IPO Explorer (searchable table of all IPOs with quick metrics)
- IPO Detail Page (one-page profile per IPO: issue details, timeline charts, valuation, grey market, anchor investors, related news)
- Predictive/Scoring Tool (user inputs → IPO quality score + explanation)
- Research & Whitepaper (downloadable PDFs + executive summary)
- Data Download (CSV/JSON API endpoints)

Frontend tech suggestions

- React (Next.js) or Vue;
- Embed Power BI/Tableau
- UI framework: Tailwind CSS
- Responsive design.

Backend / APIs

- Node.js (Express/Fastify) or Python (FastAPI) for APIs.

Deliverables (web):

- Production website repo with CI, README, and deployment script.
- Deployed URL (e.g., Vercel).

2) Data Collection & Engineering

Goal: Build a reproducible pipeline to collect, clean, and store IPO and macro data for 10–15 years.

Primary data sources

- DRHP and other available prospectuses.
- NSE / BSE historical listings & issue details.
- Financial databases: CMIE, Capitaline, Bloomberg, Moneycontrol, Economic Times data scrapes.
- Grey Market Premium (GMP) sources — public GMP trackers / archival blogs/exchange forums.
- Anchor investor allotment data
- Macroeconomic: RBI (rates/liquidity), MOSPI (GDP), CPI, interest rate series.
- Market indices: NIFTY 50, sectoral indices, NIFTY SME.

ETL steps

- Ingest raw CSV/HTML/API; implement scrapers where APIs are absent.
- Normalise fields (date formats, tickers, sector taxonomy).
- Reconcile duplicates, map IPO to listed ticker, capture delisting/merger events.
- Compute derived fields: listing gain, returns at 1M/6M/1Y, subscription rates, valuation multiples (using most recent financials at IPO filing).
- Store in a relational DB (Postgres) + object store for raw files. Provide versioning.

Data quality & metadata

- Impute missing values sensibly

Deliverables :

- Cleaned the dataset and data dictionary.
- API endpoints to query IPO data and aggregated stats.

3) Data Analysis & Modelling

Goal: Analytical framework to benchmark IPOs, cluster them, measure mispricing, and build a predictive scoring model.

Analytical tasks

- Descriptive analytics: counts by year/sector, subscription distributions, listing gains distribution, and long-term return distribution.
- Benchmarking: compare IPO returns to NIFTY/NIFTY-SME and sector indices over the same horizons.
- Valuation trends: aggregate P/E, P/B, EV/EBITDA at IPO across years and sectors.
- Mispricing analysis: measure initial-day deviation vs intrinsic proxies; compute Grey Market Premium correlations.
- Factor analysis: PCA/Factor models on fundamentals, subscription metrics, and market context.
- Clustering: k-means / hierarchical on features (growth proxies, profitability, subscription behaviour, promoter holding, GMP) to identify archetypes (speculative, value, high-growth).
- Predictive model/scoring: train models (XGBoost / LightGBM / logistic regression) to predict 1Y positive return or classify 'value vs speculative'. Provide SHAP explanations for interpretability.

Feature engineering

- Financial ratios at filing (P/E, P/B, EBITDA margins)
- Market signals: contemporaneous index returns, liquidity metrics, and interest rate change in the prior quarter
- Subscription behavioural features: retail % subscription, GMP.

Validation

- Time-series aware CV (walk-forward).
- Evaluate: AUC/precision/recall for classification; MAE/RMSE for regression; economic backtest (strategy returns from top-scored IPOs).

Deliverables (analysis):

- Jupyter notebooks/scripts reproducing results.

4) Visualisation (Dashboard content)

Goal: Convert analyses into interactive visuals and narrative-driven insights for the website.

Key visualizations

- Time-series: number of IPOs, total raised, and median listing gain over time.
- Scatter: Issue valuation vs listing gain; P/E at IPO vs 1Y return.
- Distribution: subscription rates, GMP histogram.
- Sector heatmap: sector-wise median returns with time slider.

- IPO Detail Page: issue timeline (announcement → subscription → listing), pre/post-listing price chart, valuation changes, SHAP explanation for scoring.
- **Fully integrated dashboard pages on the website.**

Tech stacks

- Frontend: React (Next.js), Tailwind CSS
- Backend: FastAPI or Node.js + Express
- Database Management: PostgreSQL
- ML: Python, pandas, scikit-learn, XGBoost/LightGBM, SHAP.
- Deployment: Vercel for hosting (optional)

Final Deliverables

1. Website with an interactive dashboard and model tools.
2. Source code repos for frontend, backend, and ETL/analysis, with CI.
3. Cleaned dataset + data dictionary and API for downloads.
4. Predictive scoring model with API endpoints.

Guidelines to note:

- Based on this doc, for the team selection at IIT-Guwahati, the participants are requested to create a document with all details, such as:
 - Ideas to solve the PS
 - Where can you help
 - Brief solution framework and event flow according to you
 - At least one additional idea can be added to make this project better, as this is an umbrella Problem statement with many domains being catered.
 - A 6 Page document has to be submitted with the attached form: <https://forms.gle/uRfJ4j3kKTKXppvc8>
 - Deadline to submit- 2nd December EoD
 - The project will be allotted to those who have completed/submitted their MF reports